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SEMICONDUCTOR DEVICE AND EXTERNAL CONNECTION MAIN TERMINAL

Abstract

An external connection main terminal includes a P main terminal and an N main terminal. The P main terminal includes an external terminal, a horizontal portion of a planar trapezoidal shape parallel to an insulated circuit substrate, and a connection terminal extending obliquely or vertically from the horizontal portion toward the insulated circuit substrate. The N main terminal includes an external terminal, a horizontal portion of a planar trapezoidal shape parallel to the insulated circuit substrate and being located over the horizontal portion of the P main terminal so that the horizontal portions of the P and N main terminals are superimposed over each other with a predetermined gap between in a direction orthogonal to the insulated circuit substrate, and a connection terminal extending obliquely or vertically from the horizontal portion toward the insulated circuit substrate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is based upon and claims the benefit of priority of the prior Japanese Patent Application No. 2024-029288, filed on Feb. 29, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present embodiment discussed herein relates to a semiconductor device and an external connection main terminal.

2. Background of the Related Art

[0003] In a semiconductor device, a collector electrode joint portion and an emitter electrode joint portion are located at the same position in an up-and-down direction and are adjacent in a right-and-left direction at a predetermined interval (see, for example, Japanese Laid-open Patent Publication No. 2013-021107). A wiring holding portion is provided that holds a first lead frame connected to a first conductive plate and having a first wiring portion wired parallel to a substrate principal surface and a second lead frame connected to a second conductive plate and having a second wiring portion wired above the front surface of the first wiring portion with a gap therebetween so as to overlap the first wiring portion (see, for example, International Publication Pamphlet No. WO 2021/029150). There is a region where a positive-electrode-side lead frame and a negative-electrode-side lead frame overlap each other (see, for example, Japanese Laid-open Patent Publication No. 2023-000131, Japanese Laid-open Patent Publication No. 2023-014524, Japanese Laid-open Patent Publication No. 2023-144474, and International Publication Pamphlet No. WO 2018/142863).

SUMMARY OF THE INVENTION

[0004] According to one aspect, there is provided a semiconductor device including: an insulated circuit substrate group including a plurality of insulated circuit substrates that are arranged on one plane; and an external connection main terminal formed of a first main terminal and a second main terminal in an integral manner, the first main terminal including a positive external terminal connected to a positive electrode of an external power supply, a first lead frame portion having a first planar trapezoidal shape parallel to the one plane, and being connected to the positive external terminal, and a first substrate connection portion extending obliquely or vertically from an edge of the first lead frame portion toward the insulated circuit substrate group, the second main terminal placed over the first main terminal and including a negative external terminal connected to a negative electrode of the external power supply, a second lead frame portion having a second planar trapezoidal shape parallel to the one plane, and being connected to the negative external terminal, the second lead frame portion located over the first lead frame portion having a predetermined gap therebetween in a direction orthogonal to the one plane, and a second substrate connection portion extending obliquely or vertically from an edge of the second lead frame portion toward the insulated circuit substrate group.

[0005] The object and advantages of the invention will be realized and attained by means of the elements and combinations particularly pointed out in the claims.

[0006] It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory and are not restrictive of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a first plan view of a semiconductor device;
[0008] FIG. 2 is a side view of the semiconductor device;
[0009] FIG. 3 is a sectional view of the semiconductor device;
[0010] FIG. 4 illustrates a second plan view of the semiconductor device;
[0011] FIG. 5 is a third plan view of the semiconductor device;
[0012] FIG. 6 is a fourth plan view of the semiconductor device;
[0013] FIG. 7 illustrates an equivalent circuit modeling the function of the semiconductor device;
[0014] FIG. 8 illustrates the configuration of a PN main terminal according to a first reference example;
[0015] FIG. 9 illustrates the configuration of a PN main terminal according to a second reference example;
[0016] FIG. 10 illustrates the configuration of a PN main terminal according to a third reference example;
[0017] FIG. 11 illustrates the evaluation of the PN main terminals in the first, second, and third reference examples;
[0018] FIG. 12 illustrates an example of the configuration of a PN main terminal according to the present embodiment;
[0019] FIGS. 13A and 13B illustrate a first example of the extending directions of connection terminals of the PN main terminal, with FIG. 13A illustrating the extending directions of the connection terminals of the P main terminal and FIG. 13B illustrating the extending directions of the connection terminals of the N main terminal;
[0020] FIGS. 14A and 14B illustrate a second example of the extending directions of the connection terminals of the PN main terminal, with FIG. 14A illustrating the extending directions of the connection terminals of the P main terminal and FIG. 14B illustrating the extending directions of the connection terminals of the N main terminal;
[0021] FIG. 15 is a view for describing the inclination angles and connection positions of the connection terminals provided in the P main terminal;
[0022] FIG. 16 is a view for describing the inclination angles and connection positions of the connection terminals provided in the N main terminal;
[0023] FIG. 17 illustrates an example of the configuration of the PN main terminal according to the present embodiment;
[0024] FIG. 18 is a view for describing a projection of an insulator;
[0025] FIG. 19 illustrates an example of a connection area of the PN main terminal; and
[0026] FIG. 20 illustrates the evaluation of the PN main terminal according to the present embodiment.

DETAILED DESCRIPTION OF THE INVENTION

[0027] Hereinafter, an embodiment will be described with reference to the accompanying drawings. In the following description, the terms “front surface” and “upper surface” refer to an X-Y plane facing up (in the +Z direction) in a semiconductor device illustrated in drawings. Similarly, the term “up” refers to an upward direction (the +Z direction) in the semiconductor device illustrated in the drawings. The terms “rear surface” and “lower surface” refer to an X-Y plane facing down (in the -Z direction) in the semiconductor device illustrated in the drawings. Similarly, the term “down” refers to a downward direction (the -Z direction) in the semiconductor device illustrated in the drawings. The same directionality applies to the other drawings, as appropriate. The terms “front surface,” “upper surface,” “up,” “rear surface,” “lower surface,” “down,” and “side surface” are used for convenience to describe relative positional relationships, and do not

limit the technical concept of the embodiment. For example, the terms “up” and “down” are not always related to the vertical directions to the ground. That is, the “up” and “down” directions are not limited to the gravity direction.

[0028] A semiconductor device will be described with reference to FIGS. 1 to 3. FIG. 1 is a first plan view of the semiconductor device. FIG. 2 is a side view of the semiconductor device. FIG. 3 is a sectional view of the semiconductor device. In this connection, FIG. 3 is a sectional view taken along a dash-dotted line X1-X1 of FIG. 1. In addition, wires are not illustrated in FIG. 3.

[0029] The semiconductor device **10** includes a heat dissipation plate **21** serving as the rear surface thereof, and a case **22** disposed on the heat dissipation plate **21** to serve as the side surfaces thereof. In addition, in the semiconductor device **10**, components are housed in a housing space **22a** surrounded by the heat dissipation plate **21** and case **22**, and the components in the housing space **22a** are sealed by a sealing material **24**. The components housed in the housing space **22a** include insulated circuit substrates, semiconductor chips disposed on the insulated circuit substrates, wires connecting these, and others. FIG. 3 illustrates insulated circuit substrates **31** and **33** as some of the insulated circuit substrates. In addition, the semiconductor device **10** includes an external connection output terminal **41**, and an external connection main terminal **40** including a main terminal **42** (first main terminal) and a main terminal **43** (second main terminal).

[0030] The heat dissipation plate **21** is a plate-shaped member that is rectangular in plan view. The outer shape of the heat dissipation plate **21** may be one size smaller than the outer shape of the case **22**. The corners of the heat dissipation plate **21** may be rounded or chamfered. The heat dissipation plate **21** is made of a metal with high heat dissipation. Examples of the metal here include copper, aluminum, and an alloy containing at least one of these. The surface of the heat dissipation plate **21** may be plated to improve its corrosion resistance. Examples of the plating material used here include nickel, a nickel-phosphorus alloy, and a nickel-boron alloy.

[0031] The insulated circuit substrates are bonded to the front surface of the heat dissipation plate **21** via a bonding material such as solder. Each insulated circuit substrate includes an insulating plate, circuit patterns formed on the front surface of the insulating plate, and a metal plate formed on the rear surface of the insulating plate.

[0032] The insulating plate is rectangular in plan view. The corners of the insulating plate may be rounded or chamfered. The insulating plate is made of ceramics with high thermal conductivity. For example, the ceramics here contain a material such as aluminum oxide, silicon nitride, or aluminum nitride as the main component thereof.

[0033] The circuit patterns are made of a metal with high electrical conductivity. Examples of the metal here include copper, aluminum, and an alloy containing at least one of these. The surfaces of the circuit patterns may be plated to improve their corrosion resistance. Examples of the plating material used here include nickel, a nickel-phosphorus alloy, and a nickel-boron alloy. In addition, the semiconductor chips, external connection output terminal **41**, and main terminals **42** and **43** are mechanically and electrically connected to the circuit patterns where appropriate. In this connection, the configuration of the circuit patterns and semiconductor chips will be described later.

[0034] The metal plate is made of a metal with high thermal conductivity as its main component. Examples of the metal here include copper, aluminum, and an alloy containing at least one of these. The surface of the metal plate may be plated to improve its corrosion resistance. Examples of the plating material used here include nickel, a nickel-phosphorus alloy, and a nickel-boron alloy.

[0035] As each insulated circuit substrate with the above components, a direct copper bonding (DCB) substrate or an active metal brazed (AMB) substrate may be used, for example. In addition, the semiconductor chips, external connection output terminal **41**, and main terminals **42** and **43** are bonded to the circuit patterns of the insulated circuit substrates via solder. Alternatively, the external connection output terminal **41** and main terminals **42** and **43** may be joined to the circuit patterns of the insulated circuit substrates by laser welding or ultrasonic welding. Lead-free solder

is used as the solder. For example, the lead-free solder contains, as its main component, at least one of a tin-silver-copper alloy, a tin-zinc-bismuth alloy, a tin-copper alloy, and a tin-silver-indium-bismuth alloy. Instead of the solder, a sintered metal may be used. The material of the sintered metal is silver, gold, nickel, copper, or an alloy containing at least one of these.

[0036] In addition, for the insulated circuit substrates having the semiconductor chips disposed thereon, wires are used to make mechanical and electrical connections between the semiconductor chips, between the semiconductor chips and the circuit patterns, and between the plurality of circuit patterns. The wires are made of a material with high electrical conductivity. Examples of the material here include gold, silver, copper, aluminum, and an alloy containing at least one of these. For example, wires with a diameter of 20 μm to 300 μm , inclusive, are used for the control electrodes of the semiconductor chips. In addition, for example, wires with a diameter of 350 μm to 500 μm , inclusive, are used to connect to the main electrodes of the semiconductor chips for main current wiring.

[0037] The semiconductor chips may be made of silicon as its main component. The semiconductor chips may include reverse conducting-insulated gate bipolar transistors (RC-IGBTs), which have both an IGBT function and a free-wheeling diode (FWD) function. A semiconductor chip of this type has a collector electrode as an input electrode on the rear surface thereof, and has a gate electrode as a control electrode and an emitter electrode as an output electrode on the front surface thereof.

[0038] The present embodiment describes the case where the semiconductor chips are RC-IGBTs, as an example. Note, however, that the semiconductor chips may be made of silicon carbide as its main component. In this case, the semiconductor chips may be, for example, power metal-oxide-semiconductor field-effect transistors (MOSFETs), and each have an FWD together with a power MOSFET. A semiconductor chip of this type has a drain electrode as an input electrode on the rear surface thereof, and has a gate electrode as a control electrode and a source electrode as an output electrode on the front surface thereof. The semiconductor chips are bonded to predetermined circuit patterns of the insulated circuit substrates via a bonding material. The bonding material may be solder, as described earlier, or a sintered metal. For example, the sintered metal may contain aluminum, copper, or an alloy containing at least one of these as its main component.

[0039] The case **22** includes a side wall portion **23** and a lid **25** covering the lower part of the side wall portion **23**. FIG. **1** is a plan view without the lid **25**. The inner surfaces of the side wall portion **23** form a rectangle in plan view. The side wall portion **23** includes first to fourth inner wall surfaces **23a** to **23d** that surround the four sides of the housing space **22a**. The first and third inner wall surfaces **23a** and **23c** correspond to the short sides, whereas the second and fourth inner wall surfaces **23b** and **23d** correspond to the long sides. The lower surface of the side wall portion **23** is bonded to the outer edge of the heat dissipation plate **21** by an adhesive or the like.

[0040] The case **22** is formed using a thermoplastic resin by integrally insert-molding the external connection output terminal **41** and the main terminals **42** and **43** into the side wall portion **23** and the lid **25**. Examples of the resin here include a polyphenylene sulfide resin, a polybutylene terephthalate resin, a polybutylene succinate resin, a polyamide resin, and an acrylonitrile butadiene styrene resin.

[0041] The housing space **22a** surrounded by the heat dissipation plate **21** and the side wall portion **23** of the case **22** is filled with the sealing material **24**. Thereby, the components disposed in the housing space **22a**, including the insulated circuit substrates, circuit patterns, semiconductor chips, wires, and others are sealed by the sealing material **24**. The sealing material is an insulating polymer gel, and preferably contains a silicone gel as its main component.

[0042] FIG. **4** illustrates a second plan view of the semiconductor device. FIG. **5** is a third plan view of the semiconductor device. In this connection, FIG. **4** is the first plan view of FIG. **1** with the sealing material **24** in the housing space **22a** of the case **22** removed. The wires are also not illustrated in FIG. **4**. FIG. **5** is the second plan view of FIG. **4** with the main terminal **43** removed.

The following describes the external connection output terminal **41** and main terminals **42** and **43** with reference to FIGS. 3 to 5.

[0043] The external connection output terminal **41** and main terminals **42** and **43** electrically connect the circuit patterns and external devices. Each of the external connection output terminal **41** and main terminals **42** and **43** is formed of a flat plate-shaped electrically conductive member. In this connection, a part of the main terminal **42** and a part of the main terminal **43** are integrally held by a terminal holding portion **44**.

[0044] One end of the external connection output terminal **41** is exposed from the upper surface of the lid **25** of the case **22** and serves as an external output terminal **41a** that connects to an external device. The external connection output terminal **41** includes the external output terminal **41a**, a horizontal portion **41b**, and a horizontal portion **41c**, arranged in this order from the one end toward the other. The horizontal portions **41b** and **41c** are horizontal to the bottom surface of the housing space **22a**. The height of the horizontal portion **41b** is lower than that of the external output terminal **41a**, and the height of the horizontal portion **41c** is lower than that of the horizontal portion **41b**. For example, the external output terminal **41a** and horizontal portions **41b** and **41c** are formed by bending the flat plate-shaped external connection output terminal **41** such that the external output terminal **41a** and the horizontal portion **41b** are connected by a vertically extending connecting portion and the horizontal portions **41b** and **41c** are likewise connected by a vertically extending connecting portion.

[0045] The external output terminal **41a** is divided into external output terminals **41a1** to **41a3** in the horizontal direction (X direction). In addition, the horizontal portion **41c** is divided into horizontal portions **41c1** and **41c2** in the horizontal direction (X direction) with a central slit **41c3** extending therebetween. Furthermore, connection terminals **41d1** and **41d2** are formed at an edge of the horizontal portion **41c1** to extend downward therefrom. The connection terminals **41d1** and **41d2** are electrically and mechanically connected respectively to circuit patterns **31b** and **31c** (to be described later) via solder. In addition, connection terminals **41d3** and **41d4** are formed at an edge of the horizontal portion **41c2** to extend downward therefrom. The connection terminals **41d3** and **41d4** are electrically and mechanically connected respectively to circuit patterns **32b** and **32c** (to be described later) via solder. Alternatively, they may be directly connected by laser welding or ultrasonic welding, for example.

[0046] On the other hand, one end of the main terminal **42** is exposed from the upper surface of the lid **25** of the case **22** and serves as an external terminal **42a** that connects to an external device (external power supply). The main terminal **42** includes the external terminal **42a**, a horizontal portion **42b**, and a horizontal portion **42c**, arranged in this order from the one end toward the other. The horizontal portions **42b** and **42c** are horizontal to the bottom surface of the housing space **22a**. The height of the horizontal portion **42b** is lower than that of the external terminal **42a**, and the height of the horizontal portion **42c** is lower than that of the horizontal portion **42b**. The external terminal **42a** and horizontal portions **42b** and **42c** are formed by bending the flat plate-shaped main terminal **42** such that the external terminal **42a** and the horizontal portion **42b** are connected by a vertically extending connecting portion and the horizontal portions **42b** and **42c** are likewise connected by a vertically extending connecting portion.

[0047] The external terminal **42a** is divided into external terminals **42a1** and **42a2** in the horizontal direction (X direction). Furthermore, connection terminals **42d1** and **42d2** are formed at edges of the horizontal portion **42c** to extend downward therefrom. The connection terminals **42d1** and **42d2** are electrically and mechanically connected respectively to circuit patterns **31a** and **32a** (to be described later) via solder. Alternatively, they may be directly connected by laser welding or ultrasonic welding, for example.

[0048] One end of the main terminal **43** is exposed above the upper surface of the lid **25** of the case **22** and serves as an external terminal **43a** that connects to the external device. The external terminal **43a** is divided into external terminals **43a1** and **43a2** in the horizontal direction (X

direction). In addition, the main terminal **43** includes the external terminal **43a** and a horizontal portion **43c**, arranged in this order from the one end toward the other. The horizontal portion **43c** is horizontal to the bottom surface of the housing space **22a**. The height of the horizontal portion **43c** is lower than that of the external terminal **43a**. The external terminal **43a** and horizontal portion **43c** are formed by bending the flat plate-shaped main terminal **43** such that the external terminal **43a** and the horizontal portion **43c** are connected by a vertically extending connecting portion. [0049] In addition, connection terminals **43d1** and **43d2** are formed at edges of the horizontal portion **43c** to extend downward therefrom. The connection terminals **43d1** and **43d2** are electrically and mechanically connected respectively to circuit patterns **33b** and **34b** (to be described later) via solder. Alternatively, they may be directly connected by laser welding or ultrasonic welding, for example.

[0050] The terminal holding portion **44** seals the connecting portion that connects the external terminal **43a** and horizontal portion **43c** of the main terminal **43**, the connecting portion that connect the external terminal **42a** and horizontal portion **42b** of the main terminal **42**, the connecting portion that connect the horizontal portions **42b** and **42c** of the main terminal **42**, and the horizontal portion **42b**. This sealing makes it possible to maintain insulation even in the case where the main terminals **42** and **43** are placed close to each other. The terminal holding portion **44** may be made of the same material as the case **22**.

[0051] Although not illustrated, a control terminal may also be provided. In this case, one end of the control terminal is exposed from the case **22** or the lid **25** of the case **22**, and the other end thereof is positioned inside the housing space **22a**. The other end of the control terminal is electrically connected to the control electrode of a semiconductor chip. The other end of the control terminal may be mechanically and electrically connected to the control electrode of the semiconductor chip with a control wire. The other end of the control terminal and the control wire may be mechanically and electrically connected via a circuit pattern. In addition, the diameter of the control wire may be smaller than that of a wire used for main current wiring.

[0052] The horizontal portion **41c** of the external connection output terminal **41**, the horizontal portion **42c** of the main terminal **42**, and the horizontal portion **43c** of the main terminal **43** are placed below the upper surface **24a** of the sealing material **24** and are therefore sealed by the sealing material **24**.

[0053] FIG. **6** is a fourth plan view of the semiconductor device. In this connection, FIG. **6** is the third plan view of FIG. **5** with the external connection output terminal **41** and main terminal **42** removed. The following describes the configuration of the insulated circuit substrates, semiconductor chips, and wires with reference to FIG. **6**.

[0054] An insulated circuit substrate group is bonded to the front surface of the heat dissipation plate **21**. The insulated circuit substrate group includes an insulated circuit substrate **31** (first insulated circuit substrate), an insulated circuit substrate **32** (second insulated circuit substrate), an insulated circuit substrate **33** (third insulated circuit substrate), and an insulated circuit substrate **34** (fourth insulated circuit substrate). Auxiliary substrates **61** and **62** are further bonded to the front surface of the heat dissipation plate **21**. The auxiliary substrate **61** is provided in the vicinity of the short side of the device located in the +Y direction with respect to the insulated circuit substrates **31** and **32**, and a lead to be connected to the insulated circuit substrates **31** and **32** is formed by patterning. In addition, the auxiliary substrate **62** is provided in the vicinity of the short side of the device located in the -Y direction with respect to the insulated circuit substrates **33** and **34**, and a lead to be connected to the insulated circuit substrates **33** and **34** is formed by patterning. In this connection, each auxiliary substrate **61** and **62** may include a circuit pattern that is connected to a control terminal to control inputs to the gate electrodes of semiconductor chips, a circuit pattern that is connected to an auxiliary emitter terminal, and others, for example.

[0055] Circuit patterns **31a** to **31c** are formed on the front surface of an insulating plate **31d** included in the insulated circuit substrate **31**. Circuit patterns **32a** to **32c** are formed on the front

surface of an insulating plate **32d** included in the insulated circuit substrate **32**. Circuit patterns **33a** and **33b** are formed on the front surface of an insulating plate **33d** included in the insulated circuit substrate **33**. Circuit patterns **34a** and **34b** are formed on the front surface of an insulating plate **34d** included in the insulated circuit substrate **34**.

[0056] The circuit pattern **31a** is T-shaped in plan view. The circuit pattern **31a** is provided close to the end in the +Y direction of the insulating plate **31d**. Semiconductor chips **51a** to **51c** are disposed on the circuit pattern **31a**. The semiconductor chips **51a** and **51b** are disposed on opposite sides in the +X directions on the circuit pattern **31a**. The semiconductor chip **51c** is disposed in the middle between the semiconductor chips **51a** and **51b** on the circuit pattern **31a**. The rear surfaces of the semiconductor chips **51a** to **51c** are mechanically and electrically bonded to the circuit pattern **31a** via solder, which is a bonding material.

[0057] The circuit patterns **31b** and **31c** are rectangular in plan view. The circuit patterns **31b** and **31c** are disposed on both sides of the projecting part of the circuit pattern **31a** projecting in the -Y direction, on the insulating plate **31d**. The output electrode on the front surface of the semiconductor chip **51a** and the circuit pattern **31b** are mechanically and electrically connected to each other with wires **111**. The output electrode on the front surface of the semiconductor chip **51b** and the circuit pattern **31c** are mechanically and electrically connected to each other with wires **112**. The output electrode on the front surface of the semiconductor chip **51c** and each of the circuit patterns **31b** and **31c** are mechanically and electrically connected to each other with wires **113**.

[0058] The circuit patterns **31b** and **33a** are mechanically and electrically connected to each other with wires **151**. The circuit patterns **31c** and **33a** are mechanically and electrically connected to each other with wires **152**.

[0059] In addition, the connection terminal **42d1** of the main terminal **42** is electrically and mechanically connected to a region **31a1** of the circuit pattern **31a** by solder, ultrasonic bonding, or another. The connection terminal **41d1** of the external connection output terminal **41** is electrically and mechanically connected to a region **31b1** of the circuit pattern **31b** by solder. The connection terminal **41d2** of the external connection output terminal **41** is electrically and mechanically connected to a region **31c1** of the circuit pattern **31c** by solder.

[0060] The circuit pattern **32a** is T-shaped in plan view. The circuit pattern **32a** is provided close to the end in the +Y direction of the insulating plate **32d**. Semiconductor chips **52a** to **52c** are disposed on the circuit pattern **32a**. The semiconductor chips **52a** and **52b** are disposed on opposite sides in the +X directions on the circuit pattern **32a**. The semiconductor chip **52c** is disposed in the middle between the semiconductor chips **52a** and **52b** on the circuit pattern **32a**. The rear surfaces of the semiconductor chips **52a** to **52c** are mechanically and electrically bonded to the circuit pattern **32a** via solder, which is a bonding material.

[0061] The circuit patterns **32b** and **32c** are rectangular in plan view. The circuit patterns **32b** and **32c** are disposed on both sides of the projecting part of the circuit pattern **32a** projecting in the -Y direction, on the insulating plate **32d**. The output electrode on the front surface of the semiconductor chip **52a** and the circuit pattern **32b** are mechanically and electrically connected to each other with wires **121**. The output electrode on the front surface of the semiconductor chip **52b** and the circuit pattern **32c** are mechanically and electrically connected to each other with wires **122**. The output electrode on the front surface of the semiconductor chip **52c** and each of the circuit patterns **32b** and **32c** are mechanically and electrically connected to each other with wires **123**.

[0062] The circuit patterns **32b** and **34a** are mechanically and electrically connected to each other with wires **153**. The circuit patterns **32c** and **34a** are mechanically and electrically connected to each other with wires **154**.

[0063] In addition, the connection terminal **42d2** of the main terminal **42** is electrically and mechanically connected to a region **32a1** of the circuit pattern **32a** by solder, ultrasonic bonding, or another. The connection terminal **41d3** of the external connection output terminal **41** is electrically and mechanically connected to a region **32b1** of the circuit pattern **32b** by solder. The connection

terminal **41d4** of the external connection output terminal **41** is electrically and mechanically connected to a region **32c1** of the circuit pattern **32c** by solder.

[0064] The circuit pattern **33a** is U-shaped in plan view. The circuit pattern **33a** is provided close to the end in the $-Y$ direction of the insulating plate **33d**. Semiconductor chips **53a** to **53c** are disposed on the circuit pattern **33a**. The semiconductor chips **53a** and **53b** are disposed on opposite sides of the circuit pattern **33a**, with the recess of the circuit pattern **33a** therebetween. The semiconductor chip **53c** is disposed in the center close to the end in $-Y$ direction of the circuit pattern **33a** between the semiconductor chips **53a** and **53b**. The rear surfaces of the semiconductor chips **53a** to **53c** are mechanically and electrically bonded to the circuit pattern **33a** by solder, which is a bonding material.

[0065] The circuit pattern **33b** is T-shaped in plan view, and is formed close to the end in the $+Y$ direction of the insulating plate **33d** within the recess of the circuit pattern. The output electrode on the front surface of the **33a**. The semiconductor chip **53a** and the circuit pattern **33b** are mechanically and electrically connected to each other with wires **131**. The output electrode on the front surface of the semiconductor chip **53b** and the circuit pattern **33b** are mechanically and electrically connected to each other with wires **132**. The output electrode on the front surface of the semiconductor chip **53c** and the circuit pattern **33b** are mechanically and electrically connected to each other with wires **133**.

[0066] The circuit pattern **34a** is U-shaped in plan view. The circuit pattern **34a** is provided close to the end in the $-Y$ direction of the insulating plate **34d**. Semiconductor chips **54a** to **54c** are disposed on the circuit pattern **34a**. The semiconductor chips **54a** and **54b** are disposed on opposite sides of the circuit pattern **34a**, with the recess of the circuit pattern **34a** therebetween. The semiconductor chip **54c** is disposed in the center close to the end in the $-Y$ direction of the circuit pattern **34a** between the semiconductor chips **54a** and **54b**. The rear surfaces of the semiconductor chips **54a** to **54c** are mechanically and electrically bonded to the circuit pattern **34a** by solder.

[0067] The circuit pattern **34b** is T-shaped in plan view, and is formed close to the end in $+Y$ direction of the insulating plate **34d** within the recess of the circuit pattern **34a**. The output electrode on the front surface of the semiconductor chip **54a** and the circuit pattern **34b** are mechanically and electrically connected to each other with wires **141**. The output electrode on the front surface of the semiconductor chip **54b** and the circuit pattern **34b** are mechanically and electrically connected to each other with wires **142**. The output electrode on the front surface of the semiconductor chip **54c** and the circuit pattern **34b** are mechanically and electrically connected to each other with wires **143**.

[0068] In addition, the connection terminal **43d1** of the main terminal **43** is electrically and mechanically connected to a region **33b1** of the circuit pattern **33b** by solder, ultrasonic bonding, or another. The connection terminal **43d2** of the main terminal **43** is electrically and mechanically connected to a region **34b1** of the circuit pattern **34b** by solder, ultrasonic bonding, or another.

[0069] The above-mentioned wires **111** to **113**, **121** to **123**, **131** to **133**, **141** to **143**, and **151** to **154** are bonding wires made of a metal with high electrical conductivity as its main component. The metal here is aluminum, copper, or an alloy containing at least one of these. In the present embodiment, the wires **111** to **113**, **121** to **123**, **131** to **133**, **141** to **143**, and **151** to **154** are used for main current wiring, and for example, have a diameter of 300 μm to 500 μm , inclusive.

[0070] In this connection, the wires **111** to **113**, **121** to **123**, **131** to **133**, **141** to **143**, and **151** to **154** may be bonded to conductive portions (circuit patterns and output electrodes on the front surfaces of semiconductor chips) using a bonding device. Each wire **111** to **113**, **121** to **123**, **131** to **133**, **141** to **143**, and **151** to **154** is subjected to ultrasonic vibrations while being pressed against the corresponding conductive portion by the bonding tool of the bonding device. Through such wedge bonding, the wires **111** to **113**, **121** to **123**, **131** to **133**, **141** to **143**, and **151** to **154** are bonded to the conductive portions.

[0071] In addition, each semiconductor chip **51a** to **51c**, **52a** to **52c**, **53a** to **53c**, and **54a** to **54c** is

an RC-IGBT, which has both the IGBT and FWD functions, as described earlier. The following describes a power conversion function of the semiconductor device **10** with reference to FIG. 7. [0072] FIG. 7 illustrates an equivalent circuit modeling the function of the semiconductor device. FIG. 7 illustrates an inverter circuit formed of the semiconductor chips **51a** to **51c** and **52a** to **52c** and the semiconductor chips **53a** to **53c** and **54a** to **54c**, which each include an RC-IGBT. Each semiconductor chip **51a** to **51c** and **52a** to **52c** includes a switching element (IGBT) **M1** and a diode element (FWD) **D1**. Each semiconductor chip **53a** to **53c** and **54a** to **54c** includes a switching element (IGBT) **M2** and a diode element (FWD) **D2**.

[0073] The semiconductor device **10** forms a half-bridge circuit including an upper arm portion A and a lower arm portion B. The upper arm portion A of the semiconductor device **10** includes the wires **111** to **113**, wires **121** to **123**, semiconductor chips **51a** to **51c**, semiconductor chips **52a** to **52c**, and main terminal **42**, which are arranged on the insulated circuit substrates **31** and **32**. The upper arm portion A also includes the external connection output terminal **41**, which is arranged on the circuit patterns **31b** and **31c** of the insulated circuit substrate **31** and the circuit patterns **32b** and **32c** of the insulated circuit substrate **32**.

[0074] The lower arm portion B of the semiconductor device **10** includes the wires **131** to **133**, wires **141** to **143**, semiconductor chips **53a** to **53c**, semiconductor chips **54a** to **54c**, and main terminal **43**, which are arranged on the insulated circuit substrates **33** and **34**.

[0075] The insulated circuit substrate **31** and the insulated circuit substrate **33** are connected to each other by the wires **151** and **152**, and the insulated circuit substrate **32** and the insulated circuit substrate **34** are connected to each other by the wires **153** and **154**. By doing so, the upper arm portion A and the lower arm portion B are connected to each other. As a result, the semiconductor device **10** is able to function as the half-bridge circuit including the upper arm portion A and the lower arm portion B.

[0076] In this case, in the semiconductor device **10**, a wiring **55a** connecting a connection point P that is connected to the positive electrode of an external power supply (not illustrated) and a connection point C1 of the input electrodes (collector electrodes) on the rear surfaces of the semiconductor chips **51a** to **51c** and **52a** to **52c** corresponds to the main terminal **42** and circuit patterns **31a** and **32a**. In other words, the main terminal **42** serves as a P main terminal that forms a positive input terminal in the half-bridge circuit.

[0077] A wiring **55c** connecting a connection point M that is connected to the terminal of a load (not illustrated) and a connection point E1C2 of the output electrodes (emitter electrodes) of the semiconductor chips **51a** to **51c** and **52a** to **52c** and the input electrodes (collector electrodes) of the semiconductor chips **53a** to **53c** and **54a** to **54c** corresponds to the external connection output terminal **41**, circuit patterns **31b** and **31c**, circuit patterns **32b** and **32c**, wires **151** and **152**, wires **153** and **154**, and circuit patterns **33a** and **34a**. In other words, the external connection output terminal **41** serves as an M terminal that forms an output terminal in the half-bridge circuit.

[0078] A wiring **55b** connecting a connection point N that is connected to the negative electrode of the external power supply and a connection point E2 of the output electrodes (emitter electrodes) of the semiconductor chips **53a** to **53c** and **54a** to **54c** corresponds to the main terminal **43** and circuit patterns **33b** and **34b**. In other words, the main terminal **43** serves as an N main terminal that forms a negative input terminal in the half-bridge circuit.

[0079] Furthermore, wirings **55d** and **55e** connecting the connection points G1 and G2, to which control signals are input, to the control electrodes (gate electrodes) of the semiconductor chips **51a** to **51c** and **52a** to **52c** and the semiconductor chips **53a** to **53c** and **54a** to **54c**, respectively, correspond to control terminals, not illustrated.

[0080] The following describes a P main terminal and an N main terminal (hereinafter, the P main terminal and the N main terminal may collectively be referred to as a PN main terminal) that are external connection main terminals in reference examples, with reference to FIGS. 8 to 11.

[0081] FIG. 8 illustrates the configuration of a PN main terminal according to a first reference

example. The PN mainterminal **210** has a pair structure formed of a P main terminal **211** and an N main terminal **213** and a pair structure formed of a P main terminal **212** and an N main terminal **214**. Note that the illustration of a terminal holding portion, which maintains the insulation between the P main terminal **211** and the N main terminal **213** and the insulation between the P main terminal **212** and the N main terminal **214**, is omitted.

[0082] One end of the P main terminal **211** is exposed on the upper surface of a case and serves as an external terminal **211a** that connects to an external device. The P main terminal **211** includes the external terminal **211a**, a horizontal portion **211b**, and a horizontal portion **211c**, arranged in this order from the one end toward the other. The horizontal portions **211b** and **211c** are horizontal (in the Y direction) to the bottom surface of a device housing space. The height of the horizontal portion **211b** is lower than that of the external terminal **211a**, and the height of the horizontal portion **211c** is lower than the horizontal portion **211b**. The external terminal **211a** and horizontal portions **211b** and **211c** are formed by bending the flat plate-shaped P main terminal **211** such that the external terminal **211a** and the horizontal portion **211b** are connected by a vertically extending connecting portion and the horizontal portions **211b** and **211c** are likewise connected by a vertically extending connecting portion. In addition, connection terminals **211d1** and **211d2** are formed at an edge of the horizontal portion **211c** to extend vertically therefrom toward an insulated circuit substrate.

[0083] In addition, the N main terminal **213** is placed over the P main terminal **211**, and one end of the N main terminal **213** is exposed above the upper surface of the case and serves as an external terminal **213a** that connects to the external device. The N main terminal **213** includes the external terminal **213a** and a horizontal portion **213c**, arranged in this order from the one end toward the other. The horizontal portion **213c** is horizontal to the bottom surface of the device housing space, and the height of the horizontal portion **213c** is lower than that of the external terminal **213a**. Moreover, the horizontal portion **213c** is formed to be longer in the +Y direction than the horizontal portion **211c** of the P main terminal **211**, and is placed to form an overlapping area where the horizontal portion **213c** is superimposed on the horizontal portion **211c** with a predetermined distance therebetween (hereinafter, such an overlapping area may be referred to as a laminated portion).

[0084] The external terminal **213a** and the horizontal portion **213c** are formed by bending the flat plate-shaped N main terminal **213** such that the external terminal **213a** and the horizontal portion **213c** are connected by a vertically extending connecting portion. In addition, connection terminals **213d1** and **213d2** are formed at an edge of the horizontal portion **213c** to extend vertically therefrom toward an insulated circuit substrate.

[0085] Each of the connection terminals **211d1** and **211d2** of the P main terminal **211** and the connection terminals **213d1** and **213d2** of the N main terminal **213** is electrically and mechanically connected to a circuit pattern via solder. Alternatively, they may be directly connected by laser welding or ultrasonic welding, for example.

[0086] On the other hand, one end of the P main terminal **212** is exposed on the upper surface of the case and serves as an external terminal **212a** that connects to the external device. The P main terminal **212** includes the external terminal **212a**, a horizontal portion **212b**, and a horizontal portion **212c**, arranged in this order from the one end toward the other. The horizontal portions **212b** and **212c** are horizontal to the bottom surface of the device housing space, the height of the horizontal portion **212b** is lower than that of the external terminal **212a**, and the height of the horizontal portion **212c** is lower than that of horizontal portion **212b**. The external terminal **212a** and horizontal portions **212b** and **212c** are formed by bending the flat plate-shaped P main terminal **212** such that the external terminal **212a** and the horizontal portion **212b** are connected by a vertically extending connecting portion and the horizontal portions **212b** and **212c** are likewise connected by a vertically extending connecting portion. In addition, connection terminals **212d1** and **212d2** are formed at an edge of the horizontal portion **212c** to extend vertically therefrom

toward an insulated circuit substrate.

[0087] In addition, the N main terminal **214** is placed over the P main terminal **212**, and one end of the N main terminal **214** is exposed above the upper surface of the case and serves as an external terminal **214a** that connects to the external device. The N main terminal **214** includes the external terminal **214a** and a horizontal portion **214c**, arranged in this order from the one end toward the other. The horizontal portion **214c** is horizontal to the bottom surface of the device housing space, and the height of the horizontal portion **214c** is lower than that of the external terminal **214a**. Furthermore, the horizontal portion **214c** is formed to be longer in the +Y direction than the horizontal portion **212c** of the P main terminal **212**, and is placed to form a laminated portion where the horizontal portion **214c** is superimposed on the horizontal portion **212c** with a predetermined distance therebetween.

[0088] The external terminal **214a** and the horizontal portion **214c** are formed by bending the flat plate-shaped N main terminal **214** such that the external terminal **214a** and the horizontal portion **214c** are connected by a vertically extending connecting portion. In addition, connection terminals **214d1** and **214d2** are formed at an edge of the horizontal portion **214c** to extend vertically therefrom toward an insulated circuit substrate.

[0089] Each of the connection terminals **212d1** and **212d2** of the P main terminal **212** and the connection terminals **214d1** and **214d2** of the N main terminal **214** is electrically and mechanically connected to a circuit pattern via solder. Alternatively, they may be directly connected by laser welding or ultrasonic welding, for example.

[0090] FIG. **9** illustrates the configuration of a PN main terminal according to a second reference example. The PN main terminal **220** includes a P main terminal **221** and an N main terminal **222**. Note that the illustration of a terminal holding portion, which maintains the insulation between the P main terminal **221** and the N main terminal **222**, is omitted.

[0091] One end of the P main terminal **221** is exposed on the upper surface of a case and serves as an external terminal **221a** that connects to an external device. The external terminal **221a** is divided into external terminals **221a1** and **221a2** in the horizontal direction (X direction). The P main terminal **221** includes the external terminal **221a**, a horizontal portion **221b**, and a horizontal portion **221c**, arranged in this order from the one end toward the other.

[0092] The horizontal portions **221b** and **221c** are horizontal to the bottom surface of a device housing space. The height of the horizontal portion **221b** is lower than that of the external terminal **221a**, and the height of the horizontal portion **221c** is lower than that of the horizontal portion **221b**. The external terminal **221a** and the horizontal portions **221b** and **221c** are formed by bending the flat plate-shaped P main terminal **221** such that the external terminal **221a** and the horizontal portion **221b** are connected by a vertically extending connecting portion and the horizontal portions **221b** and **221c** are likewise connected by a vertically extending connecting portion. In addition, connection terminals **221d1** and **221d2** are formed at an edge of the horizontal portion **221c** to extend vertically therefrom toward insulated circuit substrates.

[0093] In addition, the N main terminal **222** is placed over the P main terminal **221**, and one end of the N main terminal **222** is exposed above the upper surface of the case and serves as an external terminal **222a** that connects to the external device. The external terminal **222a** is divided into external terminals **222a1** and **222a2** in the horizontal direction. The N main terminal **222** includes the external terminal **222a** and a horizontal portion **222c**, arranged in this order from the one end toward the other. The horizontal portion **222c** is horizontal to the bottom surface of the device housing space, and the height of the horizontal portion **222c** is lower than that of the external terminal **222a** but higher than that of the horizontal portion **221c** of the P main terminal **221**. Furthermore, the horizontal portion **222c** is formed to be shorter in the -Y direction than the horizontal portion **221c** of the P main terminal **221**, and is placed to form a laminated portion where the horizontal portion **222c** is superimposed on the horizontal portion **221c** with a predetermined distance therebetween.

[0094] The external terminal **222a** and the horizontal portion **222c** are formed by bending the flat plate-shaped N main terminal **222** such that the external terminal **222a** and the horizontal portion **222c** are connected by a vertically extending connecting portion. In addition, connection terminals **222d1** and **222d2** are formed at an edge of the horizontal portion **222c** to extend vertically therefrom toward insulated circuit substrates across the horizontal portion **221c**.

[0095] Each of the connection terminals **221d1** and **221d2** of the P main terminal **221** and the connection terminals **222d1** and **222d2** of the N main terminal **222** is electrically and mechanically connected to a circuit pattern via solder. Alternatively, they may be directly connected by laser welding or ultrasonic welding, for example.

[0096] FIG. **10** illustrates the configuration of a PN main terminal according to a third reference example. The PN main terminal **230** includes a P main terminal **231** and an N main terminal **232**. Note that the illustration of a terminal holding portion, which maintains the insulation between the P main terminal **231** and the N main terminal **232**, is omitted.

[0097] One end of the P main terminal **231** is exposed on the upper surface of a case and serves as an external terminal **231a** that connects to an external device. The external terminal **231a** is divided into external terminals **231a1** and **231a2** in the horizontal direction (X direction). In addition, the P main terminal **231** includes the external terminal **231a**, a horizontal portion **231b**, and a horizontal portion **231c**, arranged in this order from the one end toward the other. The horizontal portions **231b** and **231c** are horizontal to the bottom surface of a device housing space, the height of horizontal portion **231b** is lower than that of the external terminal **231a**, and the height of the horizontal portion **231c** is lower than that of the horizontal portion **231b**. The external terminal **231a** and the horizontal portions **231b** and **231c** are formed by bending the flat plate-shaped P main terminal **231** such that the external terminal **231a** and the horizontal portion **231b** are connected by a vertically extending connecting portion and the horizontal portions **231b** and **231c** are likewise connected by a vertically extending connecting portion. In addition, connection terminals **231d1** and **231d2** are formed at an edge of the horizontal portion **231c** to extend vertically therefrom toward insulated circuit substrates.

[0098] The N main terminal **232** is placed over the P main terminal **231**, and one end of the N main terminal **232** is exposed above the upper surface of the case and serves as an external terminal **232a** that connects to the external device. The external terminal **232a** is divided into external terminals **232a1** and **232a2** in the horizontal direction. The N main terminal **232** includes the external terminal **232a** and a horizontal portion **232c**, arranged in this order from the one end toward the other. The horizontal portion **232c** is horizontal to the bottom surface of the device housing space, and the height of the horizontal portion **232c** is lower than that of the external terminal **232a** but higher than that of the horizontal portion **231c** of the P main terminal **231**. Moreover, the horizontal portion **232c** is formed to be shorter in the -Y direction than the horizontal portion **231c** of the P main terminal **231**, and is placed to form a laminated portion where the horizontal portion **232c** is superimposed on the horizontal portion **231c** with a predetermined distance therebetween.

[0099] The external terminal **232a** and the horizontal portion **232c** are formed by bending the flat plate-shaped N main terminal **232** such that the external terminal **232a** and the horizontal portion **232c** are connected by a vertically extending connecting portion. In addition, connection terminals **232d1** and **232d2** are formed at an edge of the horizontal portion **232c** to extend vertically therefrom toward insulated circuit substrates across the horizontal portion **231c**.

[0100] Each of the connection terminals **231d1** and **231d2** of the P main terminal **231** and the connection terminals **232d1** and **232d2** of the N main terminal **232** is electrically and mechanically connected to a circuit pattern via solder. Alternatively, they may be directly connected by laser welding or ultrasonic welding, for example.

[0101] In a semiconductor device, there is parasitic inductance (L component) in a conductor portion of a wiring pattern in an insulated circuit substrate. The parasitic inductance may decrease the switching speed of the switching element included in a semiconductor chip to thereby increase

switching loss, and may cause voltage fluctuations between the power supply and the ground. Therefore, there is a demand to reduce the parasitic inductance.

[0102] In order to reduce such parasitic inductance, the PN main terminal **210** in the first reference example illustrated in FIG. **8** is configured so that the number of connection terminals that are connected to insulated circuit substrates is increased to eight. However, this approach of increasing the number of connection terminals for the connection between the PN main terminal **210** and the insulated circuit substrates to thereby reduce the parasitic inductance results in a shortage of space for mounting semiconductor chips and also makes it difficult to increase the current rating of the semiconductor device.

[0103] On the other hand, the PN main terminal **220** in the second reference example illustrated in FIG. **9** is configured so that the number of connection terminals that are connected to insulated circuit substrates is reduced to four (the number of connection terminals is reduced by 50% compared to the case of FIG. **8**). This configuration has more space for mounting semiconductor chips. However, this approach of simply reducing the number of connection terminals leads to an increase in parasitic inductance due to a change in the shapes of the connection terminals.

[0104] In addition, the PN main terminal **230** in the third reference example illustrated in FIG. **10** is configured so that the number of connection terminals that are connected to insulated circuit substrates is reduced to four, as in the case of FIG. **9**.

[0105] Moreover, the PN main terminal **230** is configured so that the laminated portion between the horizontal portion **231c** of the P main terminal **231** and the horizontal portion **232c** of the N main terminal **232** is larger in area than the laminated portion between the horizontal portion **221c** of the P main terminal **221** and the horizontal portion **222c** of the N main terminal **222** in the PN main terminal **220** illustrated in FIG. **9**. By expanding the laminated portion in this manner, the PN main terminal **230** reduces the increase in parasitic inductance caused by a change in the shapes of the connection terminals resulting from the reduction in the number of connection terminals.

[0106] However, the approach of simply expanding the laminated portion results in an insufficient inter-electrode insulation distance between the different potentials of the P main terminal **231** and the N main terminal **232**. Moreover, the approach of simply expanding the laminated portion may make it difficult to perform bonding using an assembly tool. For example, during ultrasonic bonding of the PN main terminal **230** to insulated circuit substrates, an ultrasonic bonding tool may collide with parts existing around the laminated portion, making it difficult to ultrasonically bond the connection terminals at the predetermined locations on the insulated circuit substrates. This results in poor assembly workability during device assembly.

[0107] FIG. **11** illustrates the evaluation of the PN main terminals in the first, second, and third reference examples. FIG. **11** illustrates the evaluation of the PN main terminal **210** in the first reference example, the PN main terminal **220** in the second reference example, and the PN main terminal **230** in the third reference example, in terms of the following items: parasitic inductance, insulation distance, assembly workability, and current rating increase. In FIG. **11**, a circle mark indicates that the requirement of an evaluation item is satisfied, whereas a cross mark indicates that the requirement is not satisfied. The parasitic inductance requirement is determined to be satisfied if parasitic inductance is 10 nH or less.

[0108] In the configuration of the PN main terminal **210** in the first reference example, the parasitic inductance is 9.9 nH, which satisfies the parasitic inductance requirement. The insulation distance between the P main terminal **211** and the N main terminal **213** and the insulation distance between the P main terminal **212** and the N main terminal **214** are secured, which satisfies the insulation distance requirement. There is no difficulty in assembly, which satisfies the assembly workability requirement. However, it is difficult to increase the current rating, which does not satisfy the current rating increase requirement.

[0109] In the configuration of the PN main terminal **220** in the second reference example, the parasitic inductance is 13 nH, which does not satisfy the parasitic inductance requirement. The

insulation distance between the P main terminal **221** and the N main terminal **222** is secured, which satisfies the insulation distance requirement. There is no difficulty in assembly, which satisfies the assembly workability requirement. It is possible to increase the current rating, which satisfies the current rating increase requirement.

[0110] In the configuration of the PN main terminal **230** in the third reference example, the parasitic inductance is 9.2 nH, which satisfies the parasitic inductance requirement. It is difficult to secure the insulation distance between the P main terminal **231** and the N main terminal **232**, which does not satisfy the insulation distance requirement. There is difficulty in assembly, which does not satisfy the assembly workability requirement. It is possible to increase the current rating, which satisfies the current rating increase requirement.

[0111] As described above, none of the PN main terminals **210** in the first reference example, the PN main terminals **220** in the second reference example, and the PN main terminals **230** in the third reference example satisfies the requirements of all items. Therefore, they are difficult to satisfy product specifications.

[0112] The following describes the main terminals **42** and **43** that form the external connection main terminal **40** in the present embodiment. In the following description, the main terminal **42** (first main terminal) may be referred to as the P main terminal **42**, and the main terminal **43** (second main terminal) may be referred to as the N main terminal **43**. In addition, the P main terminal **42** and the N main terminal **43** may collectively be referred to as the PN main terminal **40**.

[0113] FIG. **12** illustrates an example of the configuration of a PN main terminal according to the present embodiment. The PN main terminal **40** includes the P main terminal **42** and the N main terminal **43**. In this connection, the illustration of a terminal holding portion and insulator, which maintain the insulation between the P main terminal **42** and the N main terminal **43**, is omitted.

[0114] One end of the P main terminal **42** is exposed on the upper surface of a case and serves as the external terminal **42a** (positive external terminal) that connects to an external device (external power supply). The external terminal **42a** is divided into the external terminals **42a1** and **42a2** in the horizontal direction (X direction). In addition, the P main terminal **42** includes the external terminal **42a**, horizontal portion **42b**, and horizontal portion **42c** (first lead frame portion), arranged in this order from the one end toward the other. The horizontal portions **42b** and **42c**, which form a bus bar, are horizontal to the bottom surface of the device housing space. The height of the horizontal portion **42b** is lower than that of the external terminal **42a**, and the height of the horizontal portion **42c** is lower than that of horizontal portion **42b**.

[0115] The external terminal **42a** and horizontal portions **42b** and **42c** are formed by bending the flat plate-shaped P main terminal **42** such that the external terminal **42a** and the horizontal portion **42b** are connected by a vertically extending connecting portion and the horizontal portions **42b** and **42c** are likewise connected by a vertically extending connecting portion. In addition, a first substrate connection portion **42d** is formed at an edge of the horizontal portion **42c** to extend obliquely therefrom toward insulated circuit substrates. The first substrate connection portion **42d** includes the connection terminal **42d1** (first connection terminal) and the connection terminal **42d2** (second connection terminal).

[0116] The N main terminal **43** is placed over the P main terminal **42**, and one end of the N main terminal **43** is exposed above the upper surface of the case and serves as an external terminal **43a** (negative external terminal) that connects to the external device. The external terminal **43a** is divided into the external terminals **43a1** and **43a2** in the horizontal direction (X direction). The N main terminal **43** includes the external terminal **43a** and horizontal portion **43c** (second lead frame portion), arranged in this order from the one end toward the other. The horizontal portion **43c**, which is a bus bar, is horizontal to the bottom surface of the device housing space. The height of the horizontal portion **43c** is lower than that of the external terminal **43a** but higher than that of the horizontal portion **42c** of the P main terminal **42**. Moreover, the horizontal portion **43c** is placed to form a laminated portion where the horizontal portion **43c** is superimposed on the horizontal

portion **42c** of the P main terminal **42** with a predetermined distance therebetween.

[0117] The external terminal **43a** and horizontal portion **43c** are formed by bending the flat plate-shaped N main terminal **43** such that the external terminal **43a** and the horizontal portion **43c** are connected by a vertically extending connecting portion. In addition, a second substrate connection portion **43d** is formed at an edge of the horizontal portion **43c** to vertically extend therefrom toward insulated circuit substrates across the horizontal portion **42c**. The second substrate connection portion **43d** includes the connection terminal **43d1** (third connection terminal) and the connection terminal **43d2** (fourth connection terminal). Each of the connection terminals **42d1** and **42d2** of the P main terminal **42** and the connection terminals **43d1** and **43d2** of the N main terminal **43** is electrically and mechanically connected to a circuit pattern via solder. Alternatively, they are directly connected by laser welding or ultrasonic welding, for example.

[0118] The connection terminals **42d1** and **42d2** of the P main terminal **42** extend obliquely or vertically toward the corresponding insulated circuit substrates, and the connection terminals **43d1** and **43d2** of the N main terminal **43** extend obliquely or vertically from the horizontal portion **43c** toward the corresponding insulated circuit substrates across the horizontal portion **42c**.

[0119] FIG. **12** illustrates, as an example, a configuration in which the connection terminals **42d1** and **42d2** extend obliquely and the connection terminals **43d1** and **43d2** extend vertically. However, at least either the connection terminals **42d1** and **42d2** or the connection terminals **43d1** and **43d2** may extend obliquely toward the corresponding insulated circuit substrates.

[0120] That is, the connection terminals **43d1** and **43d2** may extend obliquely and the connection terminals **42d1** and **42d2** may extend vertically. Alternatively, the connection terminals **42d1** and **42d2** and the connection terminals **43d1** and **43d2** may all extend obliquely.

[0121] FIGS. **13A** and **13B** illustrate a first example of the extending directions of the connection terminals of the PN main terminal. FIG. **13A** illustrates the extending directions of the connection terminals of the P main terminal, and FIG. **13B** illustrates the extending directions of the connection terminals of the N main terminal. The horizontal portion **42c** of the P main terminal **42** has a first trapezoidal shape that has a first long base **42c1** connected to the external terminal **42a** and a first short base **42c2** and that narrows in width in the direction from the first long base **42c1** toward the connection terminals **42d1** and **42d2**.

[0122] Likewise, the horizontal portion **43c** of the N main terminal **43** has a second trapezoidal shape that has a second long base **43c1** connected to the external terminal **43a** and a second short base **43c2** and that narrows in width in the direction from the second long base **43c1** toward the connection terminals **43d1** and **43d2**.

[0123] Here, the connection terminal **42d1** of the P main terminal **42** extends from the horizontal portion **42c** in the +Y direction and is connected to a predetermined insulated circuit substrate. The connection terminal **42d2** of the P main terminal **42** extends from the horizontal portion **42c** in the +Y direction and is connected to a predetermined insulated circuit substrate. The connection terminal **43d1** of the N main terminal **43** extends from the horizontal portion **43c** in the -X direction and is connected to a predetermined insulated circuit substrate. The connection terminal **43d2** of the N main terminal **43** extends from the horizontal portion **43c** in the +X direction and is connected to a predetermined insulated circuit substrate.

[0124] FIGS. **14A** and **14B** illustrate a second example of the extending directions of the connection terminals of the PN main terminal. FIG. **14A** illustrates the extending directions of the connection terminals of the P main terminal, and FIG. **14B** illustrates the extending directions of the connection terminals of the N main terminal. The connection terminal **42d1** of the P main terminal **42** extends from the horizontal portion **42c** in the -X direction and is connected to a predetermined insulated circuit substrate. The connection terminal **42d2** of the P main terminal **42** extends from the horizontal portion **42c** in the +X direction and is connected to a predetermined insulated circuit substrate. The connection terminal **43d1** of the N main terminal **43** extends from the horizontal portion **43c** in the +Y direction and is connected to a predetermined insulated circuit

substrate. The connection terminal **43d2** of the N main terminal **43** extends from the horizontal portion **43c** in the +Y direction and is connected to a predetermined insulated circuit substrate. As described above, the extending directions of the connection terminals may have a pattern opposite to the extending directions of the connection terminals illustrated in FIGS. **13A** and **13B**.

[0125] FIG. **15** is a view for describing the inclination angles and connection positions of the connection terminals provided in the P main terminal. In the case where the connection terminal **42d1** of the P main terminal **42** extends obliquely toward the insulated circuit substrate **31** (first insulated circuit substrate), the angle $\theta 1$ of the connection terminal **42d1** with respect to the horizontal plane of the horizontal portion **42c** ranges from 30° to 80° , inclusive. Therefore, the connection terminal **42d1** is bonded at the angle $\theta 1$ to the circuit pattern **31a** (first circuit pattern) on the insulated circuit substrate **31**.

[0126] In addition, in the case where the connection terminal **42d2** of the P main terminal **42** extends obliquely toward the insulated circuit substrate **32** (second insulated circuit substrate), the angle $\theta 2$ of the connection terminal **42d2** with respect to the horizontal plane of the horizontal portion **42c** ranges from 30° to 80° , inclusive. Therefore, the connection terminal **42d2** is bonded at the angle $\theta 2$ to the circuit pattern **32a** (second circuit pattern) on the insulated circuit substrate **32**.

[0127] The angles $\theta 1$ and $\theta 2$ are each preferably between 45° to 60° . The angles $\theta 1$ and $\theta 2$ may be set to be equal to each other or different from each other as long as they are within the above angular range.

[0128] FIG. **16** is a view for describing the inclination angles and connection positions of the connection terminals provided in the N main terminal. In the case where the connection terminal **43d1** of the N main terminal **43** extends obliquely toward the insulated circuit substrate **33** (third insulated circuit substrate), the angle $\theta 3$ of the connection terminal **43d1** with respect to the horizontal plane of the horizontal portion **43c** ranges from 30° to 80° , inclusive. Therefore, the connection terminal **43d1** is bonded at the angle $\theta 3$ to the circuit pattern **33b** (third circuit pattern) on the insulated circuit substrate **33**.

[0129] In the case where the connection terminal **43d2** of the N main terminal **43** extends obliquely toward the insulated circuit substrate **34** (fourth insulated circuit substrate), the angle $\theta 4$ of the connection terminal **43d2** with respect to the horizontal plane of the horizontal portion **43c** ranges from 30° to 80° , inclusive. Therefore, the connection terminal **43d2** is bonded at the angle $\theta 4$ to the circuit pattern **34b** (fourth circuit pattern) on the insulated circuit substrate **34**.

[0130] In this connection, the angles $\theta 3$ and $\theta 4$ are each preferably between 45° to 60° . The angles $\theta 3$ and $\theta 4$ may be set to be equal to each other or different from each other as long as they are within the above angular range.

[0131] FIG. **17** illustrates an example of the configuration of the PN main terminal according to the present embodiment. FIG. **17** illustrates the PN main terminal **40** illustrated in FIG. **12**, together with the terminal holding portion **44** and an insulator **45** that maintain the insulation between the P main terminal **42** and the N main terminal **43**. Since the terminal holding portion **44** has already been described with reference to FIG. **3**, its description is omitted.

[0132] The insulator **45** is molded to surround the horizontal portion **42c** of the P main terminal **42** and the horizontal portion **43c** of the N main terminal **43** and to fill the space in the laminated portion that is the overlapping area where the second trapezoidal shape of the horizontal portion **43c** is superimposed on the first trapezoidal shape of the horizontal portion **42c**. The molded insulator **45** maintains the insulation in the laminated portion between the horizontal portion **42c** of the P main terminal **42** and the horizontal portion **43c** of the N main terminal **43**.

[0133] Moreover, the insulator **45** has a projection **45d1** that is positioned between the connection terminal **42d1** of the P main terminal **42** and the connection terminal **43d1** of the N main terminal **43**, and a projection **45d2** that is positioned between the connection terminal **42d2** of the P main terminal **42** and the connection terminal **43d2** of the N main terminal **43**.

[0134] The insulator **45** may be made of the same material as the terminal holding portion **44** and

the case **22**. That is, the insulator **45** is made of a thermoplastic resin. Examples of the resin here include a polyphenylene sulfide resin, a polybutylene terephthalate resin, a polybutylene succinate resin, a polyamide resin, and an acrylonitrile butadiene styrene resin.

[0135] FIG. **18** is a view for describing a projection of an insulator. The projection **45d1** is formed to project from the insulator **45** and to be positioned between the connection terminal **42d1** of the P main terminal **42** connected to the insulated circuit substrate **31** and the connection terminal **43d1** of the N main terminal **43** connected to the insulated circuit substrate **33**. The sealing material **24** is applied up to a point above the lower end Le of the projection **45d1**. That is, the lower end Le of the projection **45d1** is placed lower than the upper surface **24a** of the sealing material **24**.

[0136] FIG. **19** illustrates an example of a connection area of the PN main terminal. The semiconductor device **10** has a rectangular device shape and has a short side **L1a** (first short side) and a short side **L1b** (second short side) that are the short sides thereof, and a long side **L2a** (first long side) and a long side **L2b** (second long side) that are the long sides thereof.

[0137] Here, a first direction is defined as a $-Y$ direction from the short side **L1a** toward the short side **L1b**, and a second direction is defined as a $+X$ direction from the long side **L2a** toward the long side **L2b**. The insulated circuit substrate **33** is disposed in the first direction with respect to the insulated circuit substrate **31**, and the insulated circuit substrate **32** is disposed in the second direction with respect to the insulated circuit substrate **31**. In addition, the insulated circuit substrate **34** is disposed in the first direction with respect to the insulated circuit substrate **32** and in the second direction with respect to the insulated circuit substrate **33**.

[0138] In addition, a connection area **A1** is defined as an area that corresponds to a central part located between the middle points of the long sides of the device shape in the semiconductor device **10** and that includes the area where the insulated circuit substrates **31** and **33** face each other and the area where the insulated circuit substrates **32** and **34** face each other. The first short base **42c2** of the horizontal portion **42c** and the second short base **43c2** of the horizontal portion **43c** are located in this connection area **A1**. In addition, the first long base **42c1** of the horizontal portion **42c** and the second long base **43c1** of the horizontal portion **43c** are located in the vicinity of the short side **L1b**. In addition, a line segment **m1** (first line segment) connecting the connection terminals **42d1** and **42d2** and a line segment **m2** (second line segment) connecting the connection terminals **43d1** and **43d2** are parallel to each other.

[0139] Here, in the configuration of the PN main terminal **40** described above in the present embodiment, the PN main terminal **40** is configured so that both the horizontal portion **42c** of the P main terminal **42** and the horizontal portion **43c** of the N main terminal **43** are trapezoidal-shaped. Therefore, the laminated portion between the horizontal portion **42c** and the horizontal portion **43c** that are arranged in proximity has a trapezoidal shape with an increased area, which reduces parasitic inductance and thus enables a reduction in the switching loss of the switching elements included in semiconductor chips.

[0140] Furthermore, at least either the connection terminals **42d1** and **42d2** provided in the horizontal portion **42c** or the connection terminals **43d1** and **43d2** provided in the horizontal portion **43c** are designed to be connected at an inclined angle to insulated circuit substrates. This configuration shortens the length of the P main terminal **42** in the non-laminated portion and the length of the N main terminal **43** in the non-laminated portion, which further reduces the parasitic inductance.

[0141] In addition, the insulator **45** is molded to surround the horizontal portion **42c** and the horizontal portion **43c** and to have the projection **45d1**, which is positioned between the connection terminal **42d1** of the P main terminal **42** and the connection terminal **43d1** of the N main terminal **43**, and the projection **45d2**, which is positioned between the connection terminal **42d2** of the P main terminal **42** and the connection terminal **43d2** of the N main terminal **43**. This ensures the insulation between the opposing electrodes of the P main terminal **42** and N main terminal **43**. Furthermore, since the sealing material **24** is applied to a point above the lower ends Le of the

projections of the insulator **45**, the insulation between the opposing electrodes of the P main terminal **42** and N main terminal **43** is further enhanced.

[0142] Still further, the trapezoidal horizontal portion **42c** narrows in width toward the connection terminals **42d1** and **42d2**, which are connected to insulated circuit substrates, and the trapezoidal horizontal portion **43c** narrows in width toward the connection terminals **43d1** and **43d2**, which are connected to insulated circuit substrates. This expands the laminated portion, as well as reducing the parasitic inductance and improving the assembly workability (clearance) during device assembly.

[0143] As to the assembly workability, for example, an ultrasonic bonding tool does not collide with any part existing around the laminated portion during the ultrasonic bonding of the PN main terminal **40** to the insulated circuit substrates, which makes it possible to ultrasonically bond the connection terminals at the predetermined locations on the insulated circuit substrates. This enables easy bonding using the assembly tool.

[0144] Furthermore, the PN main terminal **40** is connected to the insulated circuit substrates via the connection terminals **42d1**, **42d2**, **43d1**, and **43d2**, without increasing the number of connections to the insulated circuit substrates. This reduces the terminal connection areas and enhances the internal layout efficiency, which makes it possible to increase the current rating.

[0145] FIG. **20** illustrates the evaluation of the PN main terminal according to the present embodiment. FIG. **20** illustrates the evaluation of the PN main terminal **40** in terms of the following items: parasitic inductance, insulation distance, assembly workability, and current rating increase.

[0146] The parasitic inductance is 9.8 nH that is below 10 nH, which satisfies the parasitic inductance requirement. The insulation distance between the P main terminal **42** and the N main terminal **43** is secured, which satisfies the insulation distance requirement (the insulation distance between the different potentials in the PN main terminal is able to handle a voltage up to 3.3 kV). There is no difficulty in assembly workability, which satisfies the assembly workability requirement. Furthermore, it is possible to increase the current rating, which satisfies the current rating increase requirement. As the PN main terminal **40** satisfies the requirements of all the items as described above, it is possible to satisfy product specifications.

[0147] According to one aspect, it is achieved to reduce parasitic inductance and accordingly reduce switching loss.

[0148] All examples and conditional language provided herein are intended for the pedagogical purposes of aiding the reader in understanding the invention and the concepts contributed by the inventor to further the art, and are not to be construed as limitations to such specifically recited examples and conditions, nor does the organization of such examples in the specification relate to a showing of the superiority and inferiority of the invention. Although one or more embodiments of the present invention have been described in detail, it should be understood that various changes, substitutions, and alterations could be made hereto without departing from the spirit and scope of the invention.

Claims

1. A semiconductor device, comprising: an insulated circuit substrate group including a plurality of insulated circuit substrates that are arranged on one plane; and an external connection main terminal formed of a first main terminal and a second main terminal in an integral manner, the first main terminal including a positive external terminal connected to a positive electrode of an external power supply, a first lead frame portion having a first planar trapezoidal shape parallel to the one plane, and being connected to the positive external terminal, and a first substrate connection portion extending obliquely or vertically from an edge of the first lead frame portion toward the insulated circuit substrate group, the second main terminal placed over the first main terminal and including

a negative external terminal connected to a negative electrode of the external power supply, a second lead frame portion having a second planar trapezoidal shape parallel to the one plane, and being connected to the negative external terminal, the second lead frame portion located over the first lead frame portion having a predetermined gap therebetween in a direction orthogonal to the one plane, and a second substrate connection portion extending obliquely or vertically from an edge of the second lead frame portion toward the insulated circuit substrate group.

2. The semiconductor device according to claim 1, wherein at least one of the first substrate connection portion or the second substrate connection portion extends obliquely toward the insulated circuit substrate group.

3. The semiconductor device according to claim 2, wherein the first substrate connection portion extends obliquely at an angle in a range of 30° to 80° with respect to the first lead frame portion, and the second substrate connection portion extends obliquely at an angle in a range of 30° to 80° with respect to the second lead frame portion.

4. The semiconductor device according to claim 1, further comprising a molded insulator surrounding the first lead frame portion and the second lead frame portion to fill the gap between the first lead frame portion and the second lead frame portion.

5. The semiconductor device according to claim 4, wherein the molded insulator includes a projection located between the first substrate connection portion and the second substrate connection portion.

6. The semiconductor device according to claim 5, further comprising a sealing material, an upper surface of the sealing material being located at a position above a lower end of the projection so that the insulated circuit substrate group is covered by the sealing material.

7. The semiconductor device according to claim 1, wherein the first trapezoidal shape of the first lead frame portion has a first long base connected to the positive external terminal and a first short base, and narrows in width in a direction from the first long base thereof toward the first short base thereof, and the second trapezoidal shape of the second lead frame portion has a second long base connected to the negative external terminal and a second short base, and narrows in width in a direction from the second long base thereof toward the second short base thereof.

8. The semiconductor device according to claim 7, wherein the semiconductor device has a rectangular substrate placement area with a first short side and a second short side that are parallel to each other and extend in a short side direction, and a first long side and a second long side that are parallel to each other and extend in a long side direction, the insulated circuit substrate group includes a first insulated circuit substrate, a second insulated circuit substrate, a third insulated circuit substrate, and a fourth insulated circuit substrate, and the first insulated circuit substrate and the second insulated circuit substrate are aligned in the short side direction and face the first short side, the third insulated circuit substrate and the fourth insulated circuit substrate are aligned in the short side direction and face the second short side, the first short base of the first lead frame portion and the second short base of the second lead frame portion are located in a connection area that is a central part of the substrate placement area located in a middle of the first and second long sides and that includes an area where the first and third insulated circuit substrates face each other and an area where the second and fourth insulated circuit substrates face each other, and the first and second long bases of the first and second lead frame portions are located in a vicinity of the second short side.

9. The semiconductor device according to claim 8, wherein the first to fourth insulated circuit substrates include first to fourth circuit patterns in the connection area, respectively, the first substrate connection portion includes a first connection terminal connected to the first circuit pattern and a second connection terminal connected to the second circuit pattern, and the second substrate connection portion includes a third connection terminal connected to the third circuit pattern and a fourth connection terminal connected to the fourth circuit pattern.

10. The semiconductor device according to claim 9, wherein a first line segment connecting a

connection point of the first connection terminal and the first circuit pattern to a connection point of the second connection terminal and the second circuit pattern is parallel to a second line segment connecting a connection point of the third connection terminal and the third circuit pattern to a connection point of the fourth connection terminal and the fourth circuit pattern.

11. An external connection main terminal, comprising: a first main terminal including a positive external terminal connectable to a positive electrode of an external power supply, a first lead frame portion of a first planar trapezoidal shape parallel to one plane, and being connected to the positive external terminal, and a first substrate connection portion extending obliquely or vertically with respect to the one plane from an edge of the first lead frame portion; and a second main terminal placed over the first main terminal, the second main terminal including a negative external terminal connectable to a negative electrode of the external power supply, a second lead frame portion of a second planar trapezoidal shape parallel to the one plane, and being connected to the negative external terminal, the second lead frame portion being located over the first lead frame portion and having a predetermined gap therebetween in a direction orthogonal to the one plane, and a second substrate connection portion extending obliquely or vertically with respect to the one plane from an edge of the second lead frame portion.

12. The external connection main terminal according to claim 11, wherein at least one of the first substrate connection portion or the second substrate connection portion extends obliquely with respect to the one plane.

13. The external connection main terminal according to claim 12, wherein the first substrate connection portion extends obliquely at an angle in a range of 30° to 80° with respect to the first lead frame portion, and the second substrate connection portion extends obliquely at an angle in a range of 30° to 80° with respect to the second lead frame portion.

14. The external connection main terminal according to claim 11, further comprising a molded insulator surrounding the first lead frame portion and the second lead frame portion to fill the gap between the first lead frame portion and the second lead frame portion.

15. The external connection main terminal according to claim 14, wherein the molded insulator includes a projection located between the first substrate connection portion and the second substrate connection portion.

16. The external connection main terminal according to claim 15, further comprising a sealing material applied to a point above a lower end of the projection.

17. The external connection main terminal according to claim 11, wherein the first trapezoidal shape of the first lead frame portion has a first long base connected to the positive external terminal and a first short base, and narrows in width in a direction from the first long base toward the first short base thereof, and the second trapezoidal shape of the second lead frame portion has a second long base connected to the negative external terminal and a second short base, and narrows in width in a direction from the second long base toward the second short base thereof.

18. The external connection main terminal according to claim 17, wherein a device including an insulated circuit substrate group in the one plane has a rectangular substrate placement area with a first short side and a second short side that are parallel to each other and extend in a short side direction, and a first long side and a second long side that are parallel to each other and extend in a long side direction, the insulated circuit substrate group includes a first insulated circuit substrate, a second insulated circuit substrate, a third insulated circuit substrate, and a fourth insulated circuit substrate, and the first insulated circuit substrate and the second insulated circuit substrate are aligned in the short side direction and face the first short side, the third insulated circuit substrate and the fourth insulated circuit substrate are aligned in the short side direction and face the second short side, the first short base of the first lead frame portion and the second short base of the second lead frame portion are located in a connection area that is a central part of the substrate placement area located in a middle of the first and second long sides and that includes an area where the first and third insulated circuit substrates face each other and an area where the second and fourth

insulated circuit substrates face each other, and the first and second long bases of the first and second lead frame portions are located in a vicinity of the second short side.

19. The external connection main terminal according to claim 18, wherein the first to fourth insulated circuit substrates include first to fourth circuit patterns in the connection area, respectively, the first substrate connection portion includes a first connection terminal connected to the first circuit pattern and a second connection terminal connected to the second circuit pattern, and the second substrate connection portion includes a third connection terminal connected to the third circuit pattern and a fourth connection terminal connected to the fourth circuit pattern.

20. The external connection main terminal according to claim 19, wherein a first line segment connecting a connection point of the first connection terminal and the first circuit pattern to a connection point of the second connection terminal and the second circuit pattern is parallel to a second line segment connecting a connection point of the third connection terminal and the third circuit pattern to a connection point of the fourth connection terminal and the fourth circuit pattern.
